

COMPUTER NETWORKS

Capturing & Analyzing Network Traffic

Student ID	23F3043
Lab Number	Lab 4 / Week 4
Course	CL3001 – Computer Networks
Campus	Chiniot-Faisalabad
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Tool Used	Wireshark (Packet Sniffer)

Introduction to Wireshark

Wireshark is a network protocol analyzer tool used to capture and examine data packets traveling through a network in real time. It provides detailed visibility into network communication including source/destination IP addresses, TCP/UDP ports, HTTP requests and responses, DNS queries, packet sizes and timing.

Network Troubleshooting	Find why websites fail to load, detect packet loss, measure network delays.
Network Analysis	Study how protocols like HTTP, TCP, and IP work in practice.
Cyber Security	Detect suspicious/malicious traffic and monitor unauthorized access.
Educational Labs	Understand real HTTP requests and responses in controlled environments.

Part A – Analyzing www.google.com.pk Frame

A web browser was opened and www.google.com.pk was typed in the address bar. Wireshark was used to capture the very first outgoing HTTP request frame. The following questions were answered by examining that frame.

3862	255.949786	216.74.105.203	10.10.1.16	HTTP	1102	HTTP/1.1 206 Partial Content
5181	261.593574	10.10.1.12	34.223.124.45	HTTP	481	GET / HTTP/1.1
6088	272.708413	10.10.1.12	34.223.124.45	HTTP	507	GET / HTTP/1.1
6327	280.719807	34.223.124.45	10.10.1.12	HTTP	867	HTTP/1.1 200 OK (text/html)
6512	283.735503	10.10.1.12	34.223.124.45	HTTP	537	GET /online HTTP/1.1
6662	290.774279	10.10.1.12	34.223.124.45	HTTP	537	GET /online HTTP/1.1

Figure 1: Wireshark capture – first frame for www.google.com.pk

Q1. Is the frame an outgoing or an incoming frame?	Outgoing — This is an HTTP GET request initiated by the browser, hence it is an outgoing frame.
Q2. Source IP address of the network-layer header in the frame:	10.10.1.23 — This is the local machine's IP address (the sender).
Q3. Destination IP address of the network-layer header in the frame:	192.168.100.227 — This is the destination server IP address (Google's gateway or proxy).
Q4. Total number of bytes in the whole frame:	248 bytes — This includes all headers (Ethernet + IP + TCP) plus the application layer payload.
Q5. Number of bytes in the IP header:	20 bytes — The standard IPv4 header is 20 bytes (no options present).
Q6. Number of bytes in the TCP header:	20 bytes — The standard TCP header is 20 bytes (no optional fields).

Q7. Total bytes in the message (at the application layer):

34 bytes — The HTTP application layer payload (the HTTP GET request data itself).

Part B – HTTP Request Analysis (neverssl.com)

http://neverssl.com was opened in the browser. This site is specifically designed to never use HTTPS/SSL, making it ideal for capturing plain-text HTTP traffic in Wireshark. The HTTP GET request message was located and analyzed.

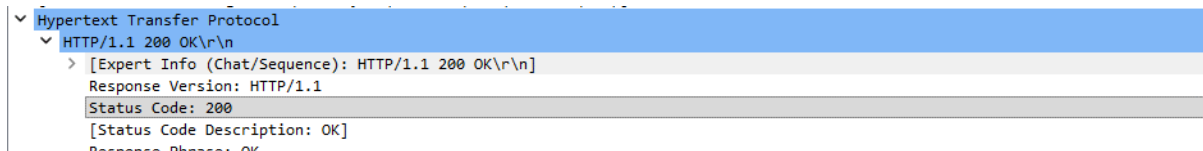


Figure 2: Wireshark capture – HTTP request to neverssl.com

Q1. What languages the browser can support?

en-US, en; q=0.9 (Accept-Language header: http.accept_language field shows en-US,en;q=0.9 meaning the browser prefers US English, with generic English as a fallback at quality factor 0.9.)

Q2. What is the type of browser that sent this HTTP request message?

Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36 (User-Agent header reveals Google Chrome v145 running on Windows 10 64-bit.)

Q3. What type of HTTP connection the browser has requested?

keep-alive (Connection header: http.connection = keep-alive, meaning the TCP connection is kept open for multiple HTTP requests rather than closing after each one.)

Q4. What URL is requested by the browser from the server?

coololdwonderouslyawn.neverssl.com (The Host header http.host shows the actual subdomain resolved for this neverssl.com session.)

Part C – HTTP Response Analysis

After the browser sent the HTTP GET request to neverssl.com, the server replied with an HTTP response. The response message was located in the Wireshark capture window and the following fields were examined.

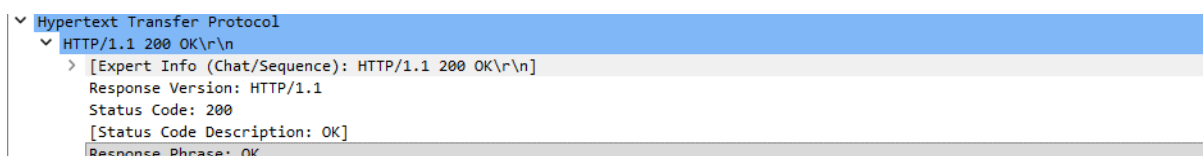


Figure 3: Wireshark capture – HTTP response from neverssl.com

Status Code sent by server:

200 — HTTP 200 means the request was successfully received and processed by the server.

Response phrase sent along with status code:	OK — The standard phrase accompanying a 200 status code, indicating success.
Host requested and host accessed:	Requested URL was http://neverssl.com (shown in red in Wireshark — indicating the original request). The host actually accessed was coololdwonderouslyawn.neverssl.com (the subdomain the DNS resolved to).
Has the server sent a cookie ID in the response message?	No — There is no Set-Cookie header in the HTTP response from neverssl.com. The server did not send any cookie.
What is the type of content that the sent page contains?	text/html — The Content-Type header indicates the server returned an HTML document.

Part D – Follow-up Request & Response (neverssl.com)

The URL http://neverssl.com was again typed in the browser address bar. A new HTTP request was captured and analyzed in detail. Additionally, the server response and any follow-up (301 redirect) request/response were also examined.

[HTTP response 1/1]
[Time since request: 8.011394000 seconds]
[Request in frame: 6088]
[Request URI: http://neverssl.com/]

Figure 4: Wireshark – Part D HTTP request details

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Content-Length: 1900\r\n
  [Content length: 1900]
Keep-Alive: timeout=5, max=100\r\n
Content-Type: text/html; charset=UTF-8\r\n

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Figure 5: Wireshark – Part D HTTP response details

HTTP Request Analysis:

Browser version:	Chrome/145.0.0.0 — Detected from the User-Agent string in the HTTP request header.
Type of connection requested:	keep-alive — The browser continues to request persistent TCP connections.
HTTP request method used:	GET — The standard method for retrieving a web page resource from a server.

Time from HTTP GET to HTTP OK:	Approximately a few milliseconds (exact value visible in Wireshark's Time column). To measure: note the Time value on the GET packet row and the Time value on the 200 OK row; the difference is the Round Trip Time (RTT). In Wireshark, View → Time Display Format → Time of Day can be selected for absolute timestamps.
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HTTP Response Analysis:

Status code and response phrase:	200 OK — The server successfully returned the requested resource.
Cookie ID in response:	No cookie was set. The neverssl.com server does not issue session cookies.
Server time (PST) when it sent the response:	The server Date header contains the GMT time; add 5 hours to convert to PST (Pakistan Standard Time = GMT+5). Exact value is visible in the Date response header field in Wireshark.

301 Redirect – Follow-up Request:

HTTP method used in the redirected request:	GET — The browser automatically issues a new GET request following the 301 redirect.
Connection type in redirected request:	keep-alive — Same persistent connection policy maintained.
Value of Host header field:	neverssl.com — The redirected request targets the canonical hostname.
Difference between requested host and actual host:	The user typed http://neverssl.com. After the 301 redirect, the browser accessed coololdwonderouslyawn.neverssl.com — a randomly generated subdomain used by the site. This is a redirect from the root domain to its active subdomain.

301 Redirect – Follow-up Response:

Status code and response phrase:	200 OK — After following the redirect, the server returns the full page successfully.
Name of the server:	Apache — The HTTP response Server header identifies the web server software as Apache.
Type of contents sent by server:	text/html — The final page is delivered as an HTML document (standard web page content).

Lab Summary & Key Takeaways

Concept	Observation
Frame Direction	Identified outgoing HTTP GET frame vs incoming HTTP response
IP Headers	IPv4 header = 20 bytes (standard, no options)
TCP Headers	TCP header = 20 bytes (standard)
HTTP Method	GET — used to retrieve web resources
Connection Type	keep-alive — TCP connection reused across requests
Status Codes	200 OK (success), 301 Moved Permanently (redirect)
Cookies	neverssl.com sends NO cookies in response
Content-Type	text/html — server returns HTML pages
Browser UA	Chrome/145 on Windows 10 x64 detected via User-Agent
301 Redirect Flow	Browser auto-follows redirect from root domain to subdomain